

REDDIT SENTIMENT ANALYSIS OF IPHONE 17

Team 12B - Andrea Pan, Angie Pang, Carson Pimental, Jaya Sruthi Raj Perikala

Table of Contents

REDDIT SENTIMENT ANALYSIS OF IPHONE 17.....	1
1. Introduction.....	2
1.1. Overview / Inspiration.....	2
1.2. Business Relevance and Research Questions.....	2
2. Data Acquisition & Description.....	3
2.1. Data Collection & Acquisition (PRAW API).....	3
2.2. Data Description and Preprocessing.....	3
3. Exploratory Data Analysis (EDA).....	3
3.1. Sentiment Distribution Overview.....	3
3.2. Engagement Metric.....	4
3.3. Engagement by Sentiment.....	4
3.4. Engagement Trends Over Time.....	4
4. Methodology.....	5
4.1. Sentiment Analysis Using VADER.....	5
4.2. Topic Modeling Using LDA.....	5
5. Results.....	7
5.1. Synopsis.....	7
5.2. Distinct Insights.....	8
5.3. Overlap.....	9
5.3.1. Battery.....	9
5.3.2. Purchase Experience.....	10
6. Recommendations and Managerial Insights.....	10
7. Bibliography.....	11

1. Introduction

1.1. Overview / Inspiration

Every new iPhone release captures significant public attention, and the launch of the iPhone 17 has continued this trend. The device quickly sparked conversations among both tech enthusiasts and casual consumers eager to share their impressions. As a group, we had our own reactions after Apple's September 9th keynote announcement, which led us to wonder how the broader public was responding to the new release. This created a timely opportunity to examine real-time sentiment surrounding the iPhone 17. Reddit emerged as the ideal platform for this analysis, as its long form posts and active tech communities allow users to share detailed reactions surrounding various topics. Unlike other social media platforms tending to favor shorter, more surface level comments, Reddit's structure encourages deeper conversations and emphasizes the most relevant perspectives through upvotes, making it the most reliable source to capture honest sentiment about the iPhone 17 in the critical weeks following its release.

1.2. Business Relevance and Research Questions

While Apple invests heavily in product development and marketing, consumer perception ultimately shapes the long-term success of each iPhone release. Understanding the themes users discuss online provides critical insight into which features resonate, which issues create frustration, and how these reactions may influence purchase decisions. Through sentiment and topic analysis, we identified patterns that highlight the aspects of the iPhone 17 that matter most to customers. These findings offer actionable guidance for Apple and stakeholders seeking to improve customer experience, and align future releases with user expectations. Our study aims to answer the following questions:

1. What core topics emerge from user discussions about the iPhone 17, and how do these themes differ across positive and negative sentiment?
2. Which product features are generating the highest levels of positive engagement, and how can Apple leverage these strengths in future product strategy?
3. Which recurring complaints or issues are contributing to negative sentiment, and what risks do they pose to customer satisfaction and overall product perception?

2. Data Acquisition & Description

2.1. Data Collection & Acquisition (PRAW API)

We collected approximately 2,600 Reddit posts using the PRAW API in Python after obtaining the required API credentials. Because PRAW API limits retrieval to 250 posts per call and does not allow direct-range specification, we first gathered 250 hot posts without restricting the subreddit and grouped them to identify which communities mentioned “iPhone 17” most often. Based on this, we selected eight subreddits with more than five mentions of “iPhone 17”: iPhone17Pro, iPhoneography, iPhone17, iPhone, Tech_Philippines, iOSsetups, patinaproud, and ATT.

2.2. Data Description and Preprocessing

After selecting the subreddits, we set “iPhone 17” as the search keyword and used five search combinations: hot posts within a year, top posts within a year, top posts within a month, relevance posts within a year, and relevance posts within a month. Using a for loop, we extracted posts from all eight subreddits.

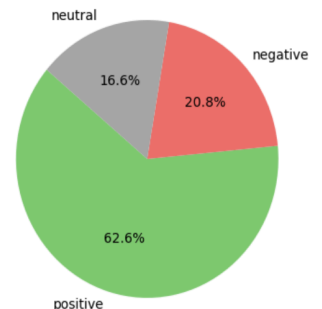
We then used `pd.concat` to combine all data frames, removed duplicates, sorted posts by creation time, and filtered for posts published after September. Posts with blank text were removed, since sentiment scoring requires textual content. During our review, we also found that most posts in the Tech_Philippines subreddit were not written in English, so we excluded that subreddit from the final dataset..

3. Exploratory Data Analysis (EDA)

3.1. Sentiment Distribution Overview

The distribution of sentiment was examined after applying VADER. The dataset consisted of 1634 positive posts, 543 negative posts, and 432 neutral posts. Positive posts formed the largest category, followed by negative sentiment, with neutral sentiment representing the smallest share of discussion.

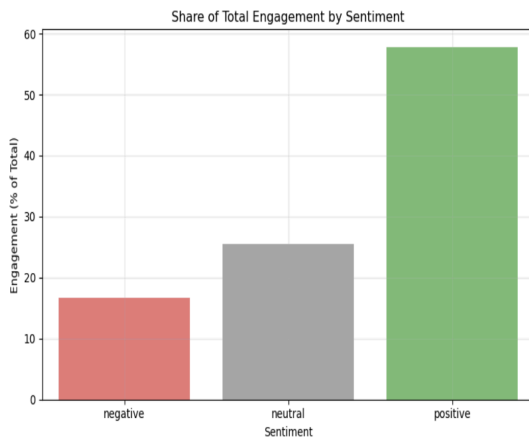
Sentiment Distribution of iPhone 17 Reddit Posts



3.2. Engagement Metric

Since we ultimately wanted to capture how actively Reddit users were expressing their opinions on different iPhone 17 themes, we created a new metric called Engagement. Engagement represents the level of user activity around iPhone 17 related discussions on Reddit, capturing how frequently posts are created and how strongly the community agrees with them (ex: upvotes), allowing us to determine which topics draw the most user activity and interaction. By summing each topic's total posts and total score value, we are able to measure not only how often a topic is discussed, but also how intensely other users interact with it. In order to ensure we were properly combining metrics with different scales we normalized the score values.

3.3. Engagement by Sentiment

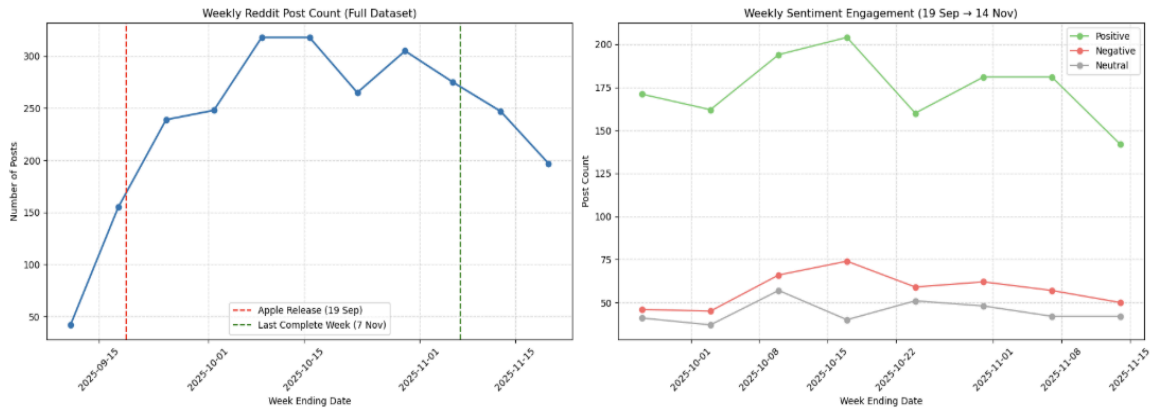


Positive sentiment drove a substantially larger share of total engagement compared to neutral and negative posts. The figure shows that positive discussions accounted for more than half of all engagement, indicating that users not only created more positive posts but also interacted with them through upvotes. Neutral sentiment captured a moderate level of engagement, while negative posts generated the least community interaction.

3.4. Engagement Trends Over Time

The following graph shows how user engagement changed over time. The left plot displays the total number of Reddit posts each week, illustrating how discussion volume rose immediately after the iPhone 17 launch and shifted in the weeks that followed. We analyzed activity from September 19, the official release date of iPhone 17, through November 14, the last fully complete week in our dataset, to ensure consistent week-to-week comparisons. The right plot breaks this activity down by sentiment and shows that positive sentiment consistently dominated post-launch engagement. Both

positive and negative posts peaked around mid-October before declining, indicating a brief rise in user reactions, including some complaints, that did not persist over time.



4. Methodology

4.1. Sentiment Analysis Using VADER

For sentiment analysis, we imported the NLTK library and SentimentIntensityAnalyzer from `nltk.sentiment.vader` in Python. VADER contains thousands of words, slang, emojis, and phrases, each labeled with a sentiment score, so we calculated each post's sentiment score using the `polarity_scores` function. After getting the score, we labeled each post based on the score. If the score is bigger than 0.05, we assigned it to the positive label; if the score is smaller than 0.05, it is labeled as negative. The remaining posts are assigned to a neutral label. VADER uses ± 0.05 as cutoffs to give the most accurate split between positive, neutral, and negative sentiment based on real-world social media text. We then added two new columns for sentiment scores and labels for further analysis.

4.2. Topic Modeling Using LDA

To gain deeper insight into the specific issues and interests driving user sentiment, we needed a model that could reveal the recurring themes within the posts themselves. Using the new dataset with labeled sentiment scores, we split our content into two dataframes one for positive sentiment text, and one for negative. For each set, we used Count Vectorizer to create a matrix, where each word from the texts is a column and each post/comment becomes a row (while ignoring English stop words). Next we conducted topic modeling by applying Latent Dirichlet Allocation (LDA) to the matrix to identify five core topics from the texts by grouping words that frequently appear together within

that sentiment. After fitting the model, we displayed the ten most impactful words for each topic in order to classify what each topic represented.¹

Negative Sentiment:

Topic 0: ['max', 've', 'issue', 'new', 'just', 'battery', 'phone', 'pro', '17', 'iphone']
 Topic 1: ['att', 'store', 'just', 'trade', 'max', 'order', 'phone', 'pro', '17', 'iphone']
 Topic 2: ['got', 'issue', 'protector', 'just', 'screen', 'phone', 'iphone', '17', 'max', 'pro']
 Topic 3: ['preview', 'https', 'redd', 'webp', 'width', 'apple', 'new', 'png', '17', 'iphone']
 Topic 4: ['like', 'just', 'issue', 'new', 'apple', 'phone', 'max', 'pro', '17', 'iphone']

Positive Sentiment:

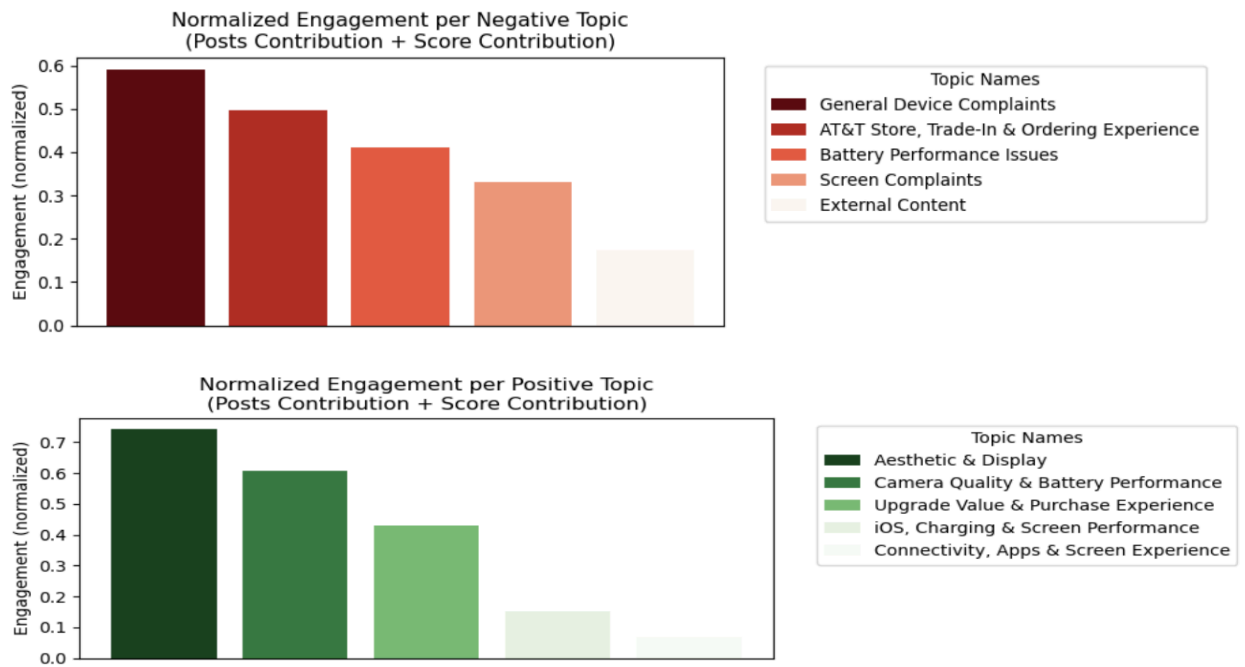
Topic 0: ['app', 'issue', 'wifi', 'data', 'screen', 'new', 'just', '17', 'iphone', 'phone']
 Topic 1: ['ve', 'screen', 'just', 'like', 'phone', 'max', 'case', 'iphone', '17', 'pro']
 Topic 2: ['screen', 'apple', '26', 'charging', 'ios', 'issue', 'max', 'pro', 'iphone', '17']
 Topic 3: ['order', 'just', 'upgrade', 'new', 'max', 'trade', 'phone', 'pro', '17', 'iphone']
 Topic 4: ['apple', 'base', 'like', 'use', 'camera', 'phone', 'battery', 'pro', '17', 'iphone']

After reviewing the top words for each topic, we assigned each topic the following general titles:

Topic	Negative	Positive
Topic 0:	Battery Performance Issues	Connectivity, Apps & Screen Experience
Topic 1:	AT&T Store, Trade-In & Ordering Experience	Aesthetic & Display
Topic 2:	Screen Complaints	iOS, Charging & Screen Performance
Topic 3:	External Content	Upgrade Value & Purchase Experience
Topic 4:	General Device Complaints	Camera Quality & Battery Performance

¹The list of words is sorted in ascending order based on their contribution to the topic, as determined by Latent Dirichlet Allocation.

Once we obtained our core topics, we compared their engagement scores to discern which themes drew the strongest interactions across our sample.



5. Results

5.1. Synopsis

Ultimately, we identified the three topics that garnered the highest engagement from each sentiment as being most significant/relevant to Apple, and ranked them accordingly.

Since our EDA proved positive engagement to largely outweigh negative engagement, we placed more emphasis on the positive topics, leading to the following rankings:

Positive

1. Aesthetic & Display
2. Camera Quality & Battery Performance
3. Upgrade Value & Purchase Experience

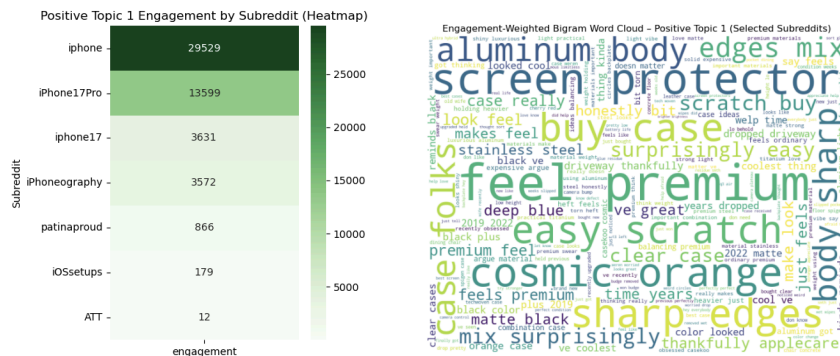
Negative

1. General Device Complaints
2. AT&T Store, Trade-In & Ordering Experience
3. Battery Performance Issues

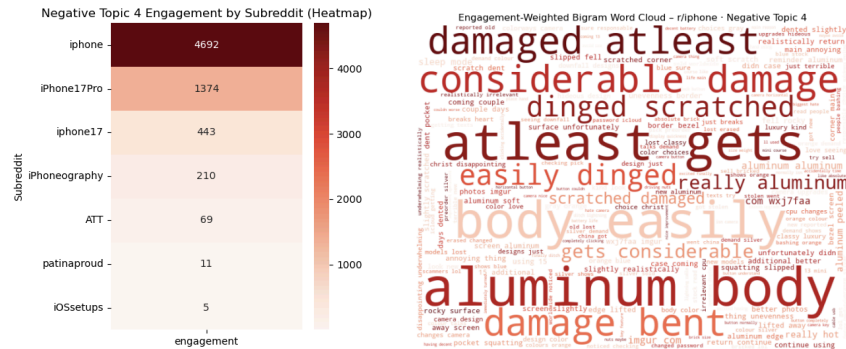
5.2. Distinct Insights

The strongest, and most relevant finding for Apple revolves around the aesthetic and display of the iPhone 17 (Insight 1). To understand which specific features users appreciated, we identified the subreddit that contributed most to this topic.

Using a word cloud to highlight the most frequently mentioned descriptors related to the positive aesthetic topic, we found that members of the ‘iPhone’ and ‘iPhone17Pro’ subreddits especially appreciated the new ‘cosmic orange’ color choice offered with the iPhone 17 Pro. Users in both communities also praised the phone’s ‘sharp edges’ and overall ‘feel’ of the premium version. This is extremely noteworthy because the iPhone 17 marks Apple’s return to more angular design elements, moving away from the rounded edges of recent generations. Apple also reintroduced an aluminum unibody design, shifting away from the titanium used in the iPhone 15 and 16 and returning to a material last seen in the iPhone 5 and 6.



Within the negative topics, we found that the ‘General Device Complaints’ theme (Insight 4) received the most interaction amongst users, making it the largest area of concern for Apple. Once again, we compared the total engagement scores of each subreddit to identify the largest contributor to this topic. After identifying ‘iPhone’, we created a word cloud in the same manner as we did for Insight 1, and found ‘iPhone’ members to voice several complaints regarding the durability of the phone including ‘easily dinged’ and ‘considerable damage’.

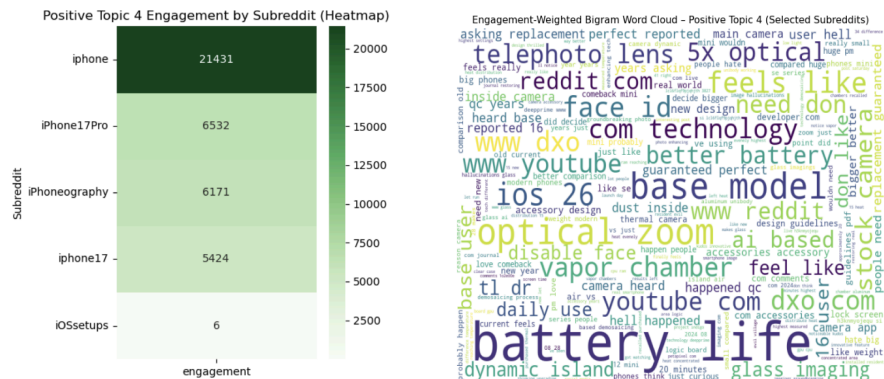


5.3. Overlap

We noticed two themes that appear in both the positive and negative sentiment analyses: battery performance (Insights 2 and 6) and purchase experience (Insights 3 and 5). To properly diagnose the customer's feelings, we needed further examination:

5.3.1. Battery

The battery's appearance in both sentiment categories reflects mixed opinions about its performance. To clarify overall perception, we summed up the total posts related to the battery topic in both positive and negative sentiments. We found that 83.72% of all battery-related engagements were of positive sentiment, while only 14.28% of engagement were of negative sentiment. This indicates that users are largely satisfied with the iPhone 17's battery performance, making the positive feedback more influential than the negative comments.



A further distinction between positive and negative buzz topics was the frequent mention of the camera in the positive sentiment, showing that people enjoy the optical zoom capabilities and are pleased with the enhancements over previous models.

5.3.2. Purchase Experience

The purchase interactions showing up in both sentiments indicate inconsistencies in Apple's customer experience, specifically regarding upgrades, trade-ins, and order processing. This reveals that some customer frustrations are stemming from issues at external retailers, and iPhone 17 sales may be indirectly suffering drawbacks from third-party partners. We found it worth noting that the AT&T store in particular is frequently mentioned by customers in negative sentiment.

6. Recommendations and Managerial Insights

Our sentiment analysis of iPhone 17 shows that positive conversations focus on the phone's aesthetics, display quality, battery performance, and camera improvements, while negative sentiment is driven by durability concerns and inconsistent purchase experiences, particularly through third-party retailers such as AT&T.

The features generating the strongest engagement are the new aesthetic design, camera quality and battery performance. Users consistently praised the cosmic orange color, sharper edges, and premium feel, indicating that Apple's design choices are resonating strongly and should be emphasized in marketing campaigns.

In contrast, the recurring complaints highlight two areas of risk that Apple should address. Reports of the device being "easily dinged" or susceptible to damage raise concerns about durability that could undermine the product's premium perception. Additionally, ongoing frustrations with upgrades, trade-ins, and order processing at third-party retailers poses a reputational risk, as customers might attribute these issues to Apple. Addressing durability and improving coordination with retail partners will help reduce negative sentiment and ensure a more consistent and reliable customer experience.

7. Bibliography

Hollington, Jesse. "The Iphone 17 pro Is Unveiled with a Powerful New Design." *iDrop News*, 9 Sept. 2025,
www.idropnews.com/news/the-iphone-17-pro-is-unveiled-with-a-powerful-new-design/252933.

Zibreg, Christian. "Thanks to Unibody Design, the Iphone 5 Is Likely the Most Durable Iphone Yet." *iDownloadBlog.Com*, 21 Sept. 2012,
www.idownloadblog.com/2012/09/21/iphone-5-drop-test/.

Shanklin, Will. "IPhone 6 vs. Iphone 6 plus: A Closer Look." *New Atlas*, 2 May 2015,
newatlas.com/iphone-6-vs-iphone-6-plus/35880.